AMRITA VISHWA VIDYAPEETHAM | School of Engineering

Embedded Systems Project

Ghost Vehicle Detection in Fog Using Pressure-Wave Sensing Before Visual Detection

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INTRODUCTION

360° MULTI-DIRECTIONAL ACOUSTIC DETECTION

Detecting approaching vehicles in dense fog using acoustic pressure waves

1

THE PROBLEM

In dense fog, low visibility and delayed detection lead to high-speed collisions.

2

THE SOLUTION

Detect approaching vehicles using acoustic pressure waves from all directions (360°).

3

USE CASES

- Foggy highways
- Blind curves
- Unsignalized rural intersections

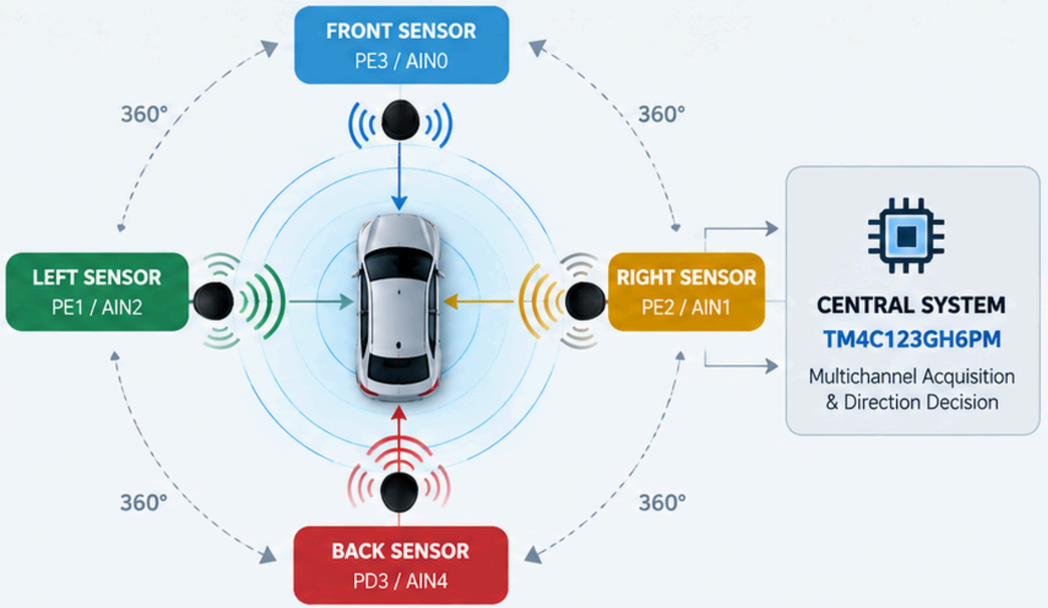
4

KEY TAKEAWAY

Listen in 360° before you see.

SYSTEM OVERVIEW

Four acoustic sensors capture sound from all directions. The central unit analyzes the signals and determines the direction of the approaching vehicle.



LITERATURE SURVEY

Study of the research work

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
1.	**Adaptive Sensor Fusion for Robust Perception in Dense Fog: A Gated Vision and LiDAR Integration Framework**	**JUNE 2026**	Improve obstacle detection in dense fog using adaptive multi-modal sensor fusion.	**Hardware:** Gated Camera, 3D LiDAR. **Software:** AFW-Net, Attention-based Adaptive Fusion, Uncertainty Estimation, Cross-modal Feature Extraction.	Achieved 82% AP at 50 m visibility, improving detection by 23.7% over conventional RGB-LiDAR fusion.	Adaptive sensor weighting, Reliable sensor fusion, Robust fog perception
2.	**Ensemble Artificial Intelligence Framework for Real-Time Automatic Vehicle Detection and Classifications**	**May 2026**	Develop an ensemble AI framework for accurate vehicle detection and classification under adverse weather.	**Hardware**: RGB Camera. **Software**: YOLOv5, YOLOv7, YOLOv8, YOLOv10, DETR-ResNet50, Faster R-CNN, Dynamic Weighting, NMS, Ensemble Learning.	Achieved 96.6% mAP, 94.2% Precision, and 94.1% F1-score for multi-class vehicle detection.	Ensemble learning, High detection accuracy, Vision-dependent
3.	**FogGate-YOLO: Traffic Object Detection in Foggy Environments Using Channel Selection Mechanisms**	**MAR 2026**	Design a lightweight object detector specifically optimized for foggy environments.	**Hardware:** RGB Camera. **Software:** YOLOv12, A2C2f-FSA Module, Bidirectional Feature Fusion Module (BFFM), GSConv, F-WIoU v3 Loss.	Improved mAP@0.5 by 5.2% (RTTS) and 7.3% (VOC-FOG) while maintaining real-time inference.	Fog-aware feature enhancement, Real-time detection, Camera limitations
4.	**FogFusion: Robust 3D object detection based on camera-LiDAR fusion for autonomous driving in foggy weather conditions**	**MAR 2026**	Improve 3D object detection in fog using enhanced camera-LiDAR fusion.	**Hardware:** RGB Camera, LiDAR. **Software:** Depth Completion Network with Fog Convolution (DCFC), Flexible Cylindrical Voxel (FCV), Cylindrical Fusion Module (CFM).	Improved detection performance by ≥3.32% over baseline methods on KITTI-C.	Enhanced 3D perception, Virtual point clouds, Costly sensor fusion
5.	**Dynamic Visibility Recognition and Driving Risk Assessment Under Rain–Fog Conditions Using Monocular Surveillance Imagery**	**JAN 2026**	Estimate road visibility and driving risk under rain-fog conditions using roadside surveillance images.	**Hardware:** Monocular CCTV Camera. **Software:** Koschmieder Scattering Model, Dark Channel Prior, Monte Carlo Simulation, Low Visibility Risk Field Force (LVRFF).	Achieved >86% visibility estimation accuracy and effectively identified high-risk driving zones.	Visibility estimation, Risk assessment, Infrastructure-based monitoring

LITERATURE SURVEY

Study of the research work

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
6.	**ECE-VDTDA: A Robust and Computationally Efficient Collision Avoidance System for Driver Assistance in Foggy Weather**	**FEB 2026**	Enhance performance in foggy-weather using a cost-effective vehicle detection system for distance, speed, and TTC-based collision warnings.	**Software:** SimYOLO-V5s_WIOU detector, Deep-SORT-based tracking **Hardware:** RGB cameras with edge-capable platforms for real-time inference	SimYOLO-V5s_WIOU improved fog detection accuracy (up to +17.45% mAP50) on foggy driving, enhanced tracking reliability, and enabled TTC-based graded collision warnings.	SimSPPF and Wise-IoU improve fog detection reliability, while robust tracking and TTC-based warnings support safer collision avoidance.
7.	**Multi-Target Tracking with Collaborative Roadside Units Under Foggy Conditions**	**JAN 2026**	Improve foggy roadside LiDAR tracking by reducing point-cloud noise and enabling collaborative multi-RSU perception.	**Hardware:** Roadside LiDAR-equipped RSUs for fog-affected traffic data collection. **Software:** Fog-noise removal using adaptive filtering and multi-RSU cooperative tracking with PHD filtering	Improved detection accuracy by +8% (thin fog) and +29% (thick fog), with enhanced multi-class tracking performance.	Noise-aware denoising and multi-RSU PHD fusion improve foggy LiDAR detection, range, and tracking stability.
8.	**Fog-YOLO: A Real-Time Object Detector for Foggy Autonomous Driving Scenarios**	**FEB 2026**	Develop a low-latency, fog-robust detector for autonomous driving with improved accuracy.	**Hardware:** RGB cameras with GPU-enabled platforms for real-time inference. **Software:** Fog-YOLO with fog-aware modules and improved detection head	Improved mAP over YOLO baseline with better small/occluded object detection in varying fog while maintaining real-time performance.	Fog-aware features and improved detection heads enhance bounding boxes and reliability for foggy perception.
9.	**3D Vision Object Detection for Autonomous Driving in Fog Using LiDAR**	**APR 2025**	Improve foggy 3D object detection using LiDAR-based perception and fog-aware processing.	**Hardware:** Automotive LiDAR sensors evaluated under foggy conditions. **Software:** LiDAR-based 3D detection with fog-mitigation techniques	Improved 3D detection accuracy and recall under fog compared to uncompensated LiDAR baselines.	Fog-aware processing preserves point-cloud features, enabling reliable 3D detection in poor visibility.
10.	**Simulation of LiDAR Under Fog with Generative Adversarial Networks for Robust 3D Object Detection**	**OCT 2025**	Generate realistic foggy LiDAR data using WGAN–LSTM to improve 3D detector robustness without real fog data collection.	**Hardware:** Velodyne VLP-32C LiDAR with fog-chamber and onboard ROS **Software:** ROS-based LiDAR processing with WGAN-GP + LSTM fog simulation	Generated 80,000 foggy LiDAR frames using WGAN–LSTM, improving 3D detector robustness and reducing the need for real fog data collection.	GAN-based fog simulation reduces real-world data dependence and improves autonomous vehicle perception robustness.

LITERATURE SURVEY

Study of the research work

S.NO	Title of the Paper	Year	Objective	Hardware /Software	Outcome	Inference
11.	**Long range vehicle detection and tracking using acoustic arrays**	**JUNE 2026**	The paper aims to develop a passive acoustic system for long-range vehicle detection and tracking, even in poor visibility and harsh outdoor conditions.	**Software:** Beamforming, Accoustic source detection, , Tracking Algorithm, PCA, Sensor Fusion. **Hardware:** accoustic sensor array, optional acoustic baffle, data acquisition/recording	Field tests demonstrated reliable tracking up to 1 km, with the acoustic baffle reducing interference and improving signal quality.	Passive acoustic arrays offer stealthy long-range tracking, with baffles improving outdoor performance. They can complement radar and optical sensors.
12.	**Distributed acoustic sensing enables real-time traffic monitoring and vehicle motion-state prediction**	**MAR 2026**	To develop a DAS-based traffic monitoring system using RNNs with spatio-temporal attention, evaluating accuracy, efficiency, interpretability, and spatial transferability.	**Hardware:** DAS Interrogator, Fiber infrastructure **Software:** Signal Processing, Feature extraction, ML/DL	The study used DAS data to classify noise, cars, and buses, achieving ~89% accuracy. Attention improved efficiency and interpretability, with good cross-location transferability.	DAS + RNNs enable continuous traffic monitoring, with attention improving accuracy and efficiency. Spatial transferability supports scalable city-wide monitoring.
13.	**Track-Before-Detect Based Multiple Vehicle Tracking with Stereo Microphone Arrays**	**APR 2026**	To track multiple vehicles in 2D using passive stereo microphones despite weak, noisy, or overlapping sounds, by jointly estimating their presence and trajectories.	**Hardware:** Multiple Microphones, Microphone Arrays **Software:** Track Before Detect, Measurement Whitening, Multi-bin angular sound-intensity processing, Labeled Multi-Bernoulli filter	The preprint presents a TBD-LMB filter with measurement whitening for multi-target acoustic tracking. Earlier work showed accurate, low-variance 2D tracking, but the abstract provides no numerical performance results.	Passive acoustic sensing can track multiple vehicles by accumulating weak signals over time. The LMB filter estimates vehicle count and tracks, useful in low-visibility conditions.
14.	**Adaptive vehicle detection in foggy conditions based on improved YOLOv5.**	**JUL 2026**	To develop an adaptive real-time vehicle detector that maintains accuracy in different fog conditions, especially for small or partially occluded vehicles.	**Hardware:** Vehicle mounted cameras, LiDAR, Radar **Software:** GANs, YOLO, Image Processing Methods	The system achieves 97.7% mAP and 33.52 FPS, outperforming YOLOv5 by 4.2%. It handles light/medium fog directly, while dense fog uses GCA-Net defogging.	Adaptive switching improves accuracy and efficiency by using direct detection in light/moderate fog and defogging in dense fog. The 33.52 FPS suggests potential for real-time use.
15.	**Autonomus driving vehicle detection methods under different low light scnes**	**APR 2026**	To examine how night, fog, and rain affect vehicle detection, review existing solutions and datasets, and identify challenges and future directions for robust autonomous-driving perception.	**Hardware:** Velodyne VLP-32C LiDAR with fog-chamber and onboard ROS **Software:** ROS-based LiDAR processing with WGAN-GP + LSTM fog simulation	The paper provides a structured review of vehicle-detection methods, datasets, challenges, and future directions, with no new model, dataset, or quantitative results.	Reliable perception requires condition-aware processing for night, fog, and rain. GAN enhancement and YOLO detectors are useful, but robustness remains challenging.

RESEARCH GAP & PROBLEM FORMULATION

- Existing studies mainly use **vision/LiDAR-based** methods; acoustic studies focus on **tracking or monitoring** with complex setups.
- There is a need for a **low-cost, real-time, 360° acoustic sensing system** to detect and identify approaching vehicles and compare **acoustic vs visual detection** for earlier warning in fog.

1 Existing Fog Detection

Vision & LiDAR systems for vehicle detection in fog.

Camera (RGB)

LiDAR (3D / Radar)

2 Sensor Dependencies

Heavy reliance on optical path or high-power active sensors.

Camera blocked

LiDAR ineffective

3 Core Limitations

Severe signal attenuation in fog, high computational overhead and costly infrastructure.

4 Research Gap

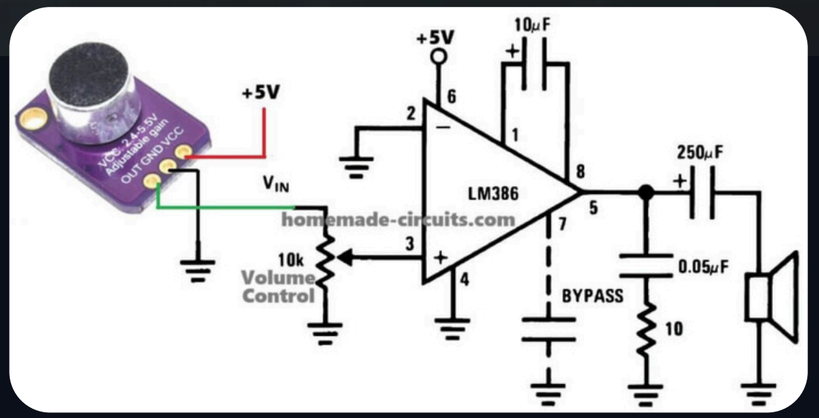
Limited exploration of 360° acoustic/pressure-wave sensing for early detection before visual signals.

5 Proposed Approach

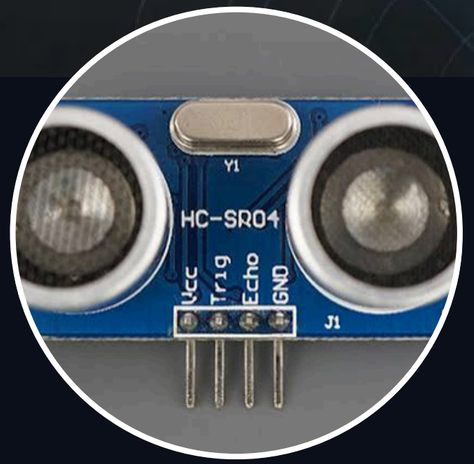
Low-cost 360° multi-sensor acoustic sensing system with TM4C123GH6PM for directional detection and early warning in fog.

HARDWARE COMPONENTS

TM4C123GH6PM

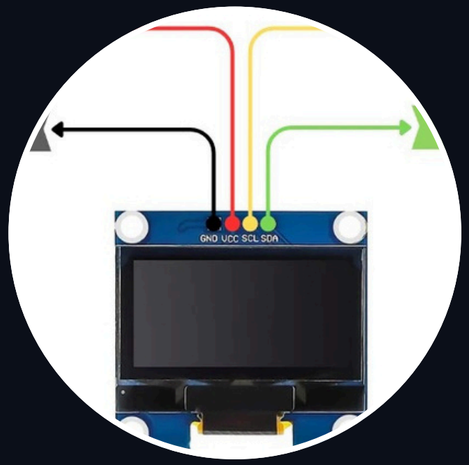


MAX4466 ANALOG MICROPHONE MODULE

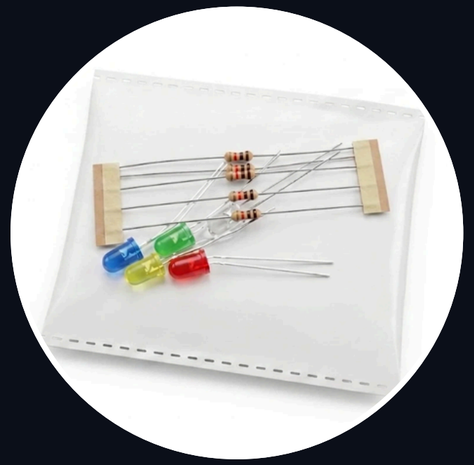


ULTRASONIC SENSOR

BUZZER

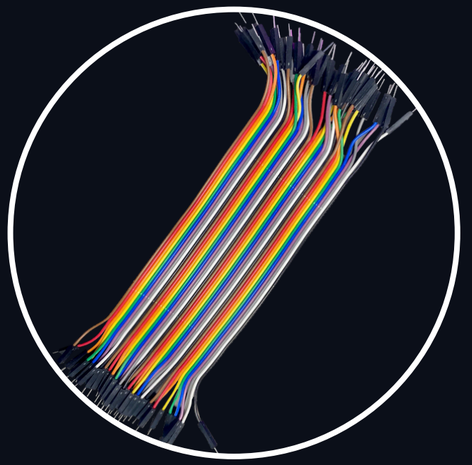


SSD1306 OLED DISPLAY



LEDS AND RESISTORS

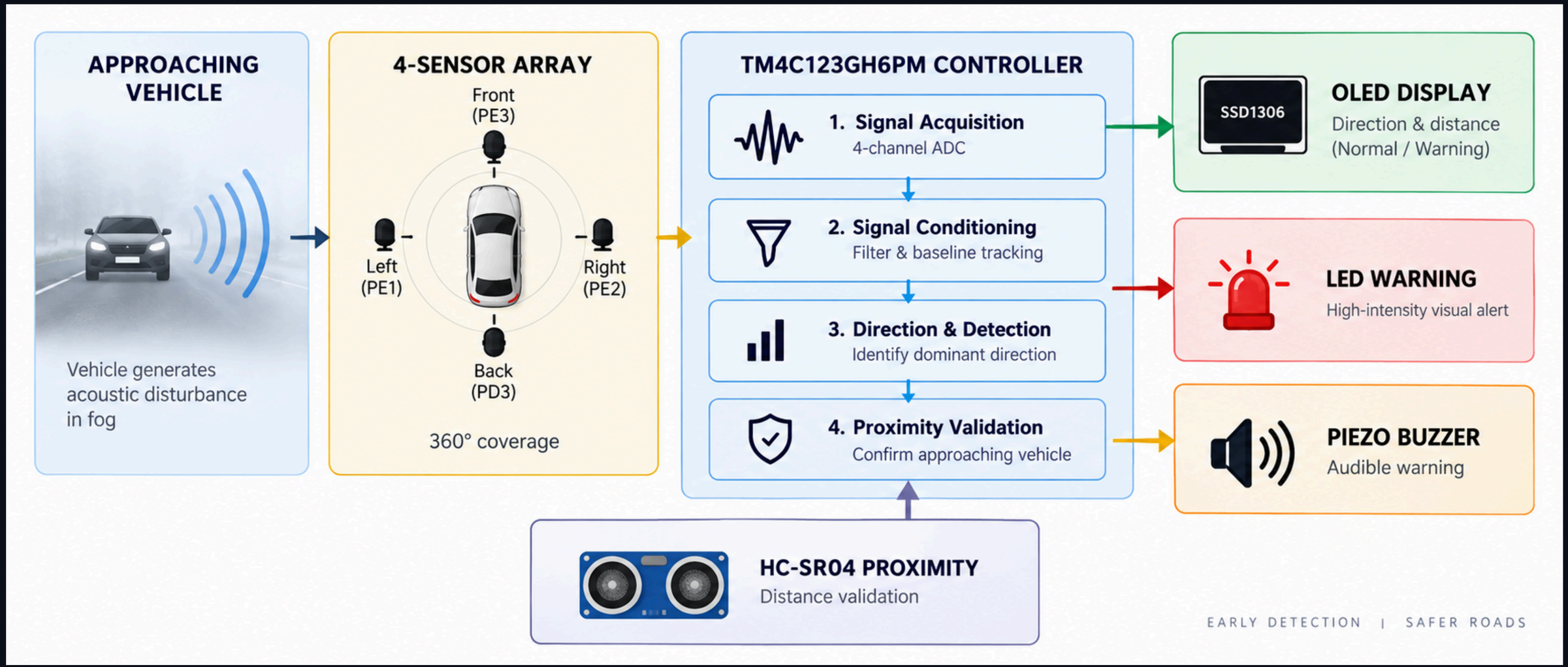
BREADBOARD



JUMPER WIRES

USB CABLE

PROPOSED SYSTEM ARCHITECTURE

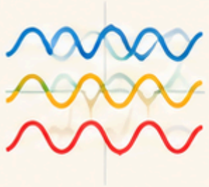


360-DEGREE SENSOR ARRANGEMENT & SPATIAL COVERAGE

1 360° SPATIAL COVERAGE

- Four sensors (Front, Right, Left, Back) placed orthogonally.
- Provides continuous 360° acoustic coverage.

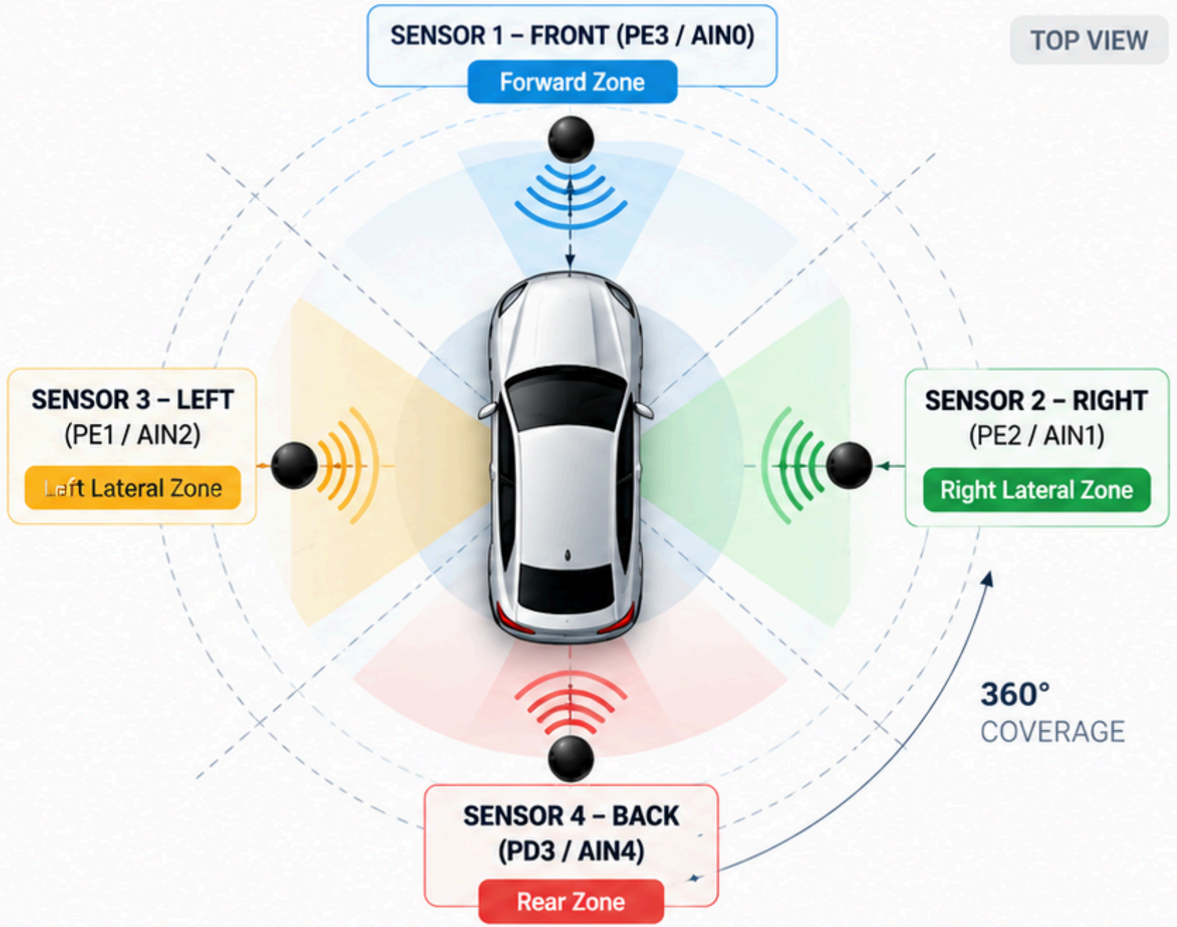
2 DIFFERENTIAL RESPONSE COMPARISON



- TM4C123 samples all 4 channels simultaneously.
- Amplitude differences indicate the approach sector.

3 DOMINANT DIRECTION ESTIMATION

- Channel with highest filtered energy is the dominant source.
- Direction output mapped to front, right, left or back region.



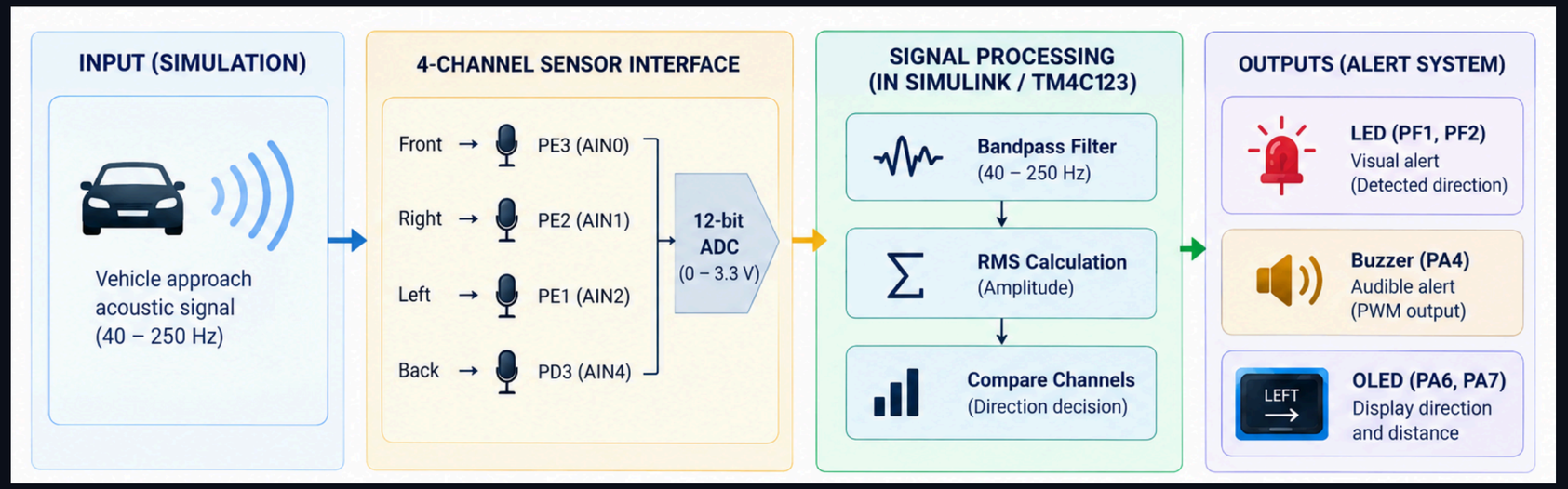
Complete 360° monitoring of approaching vehicles

Simultaneous 4-channel acquisition on TM4C123

Direction identified using amplitude comparison

Enables early detection from any direction

SIMULATION MODEL: 4-CHANNEL TM4C123 ARCHITECTURE



ADC PIN MAPPING (INPUT)

Sensor Position	TM4C123 Pin	ADC Channel	Used For
Front	PE3	AIN0	Front signal
Right	PE2	AIN1	Right signal
Left	PE1	AIN2	Left signal
Back	PD3	AIN4	Back signal

OUTPUT PIN MAPPING

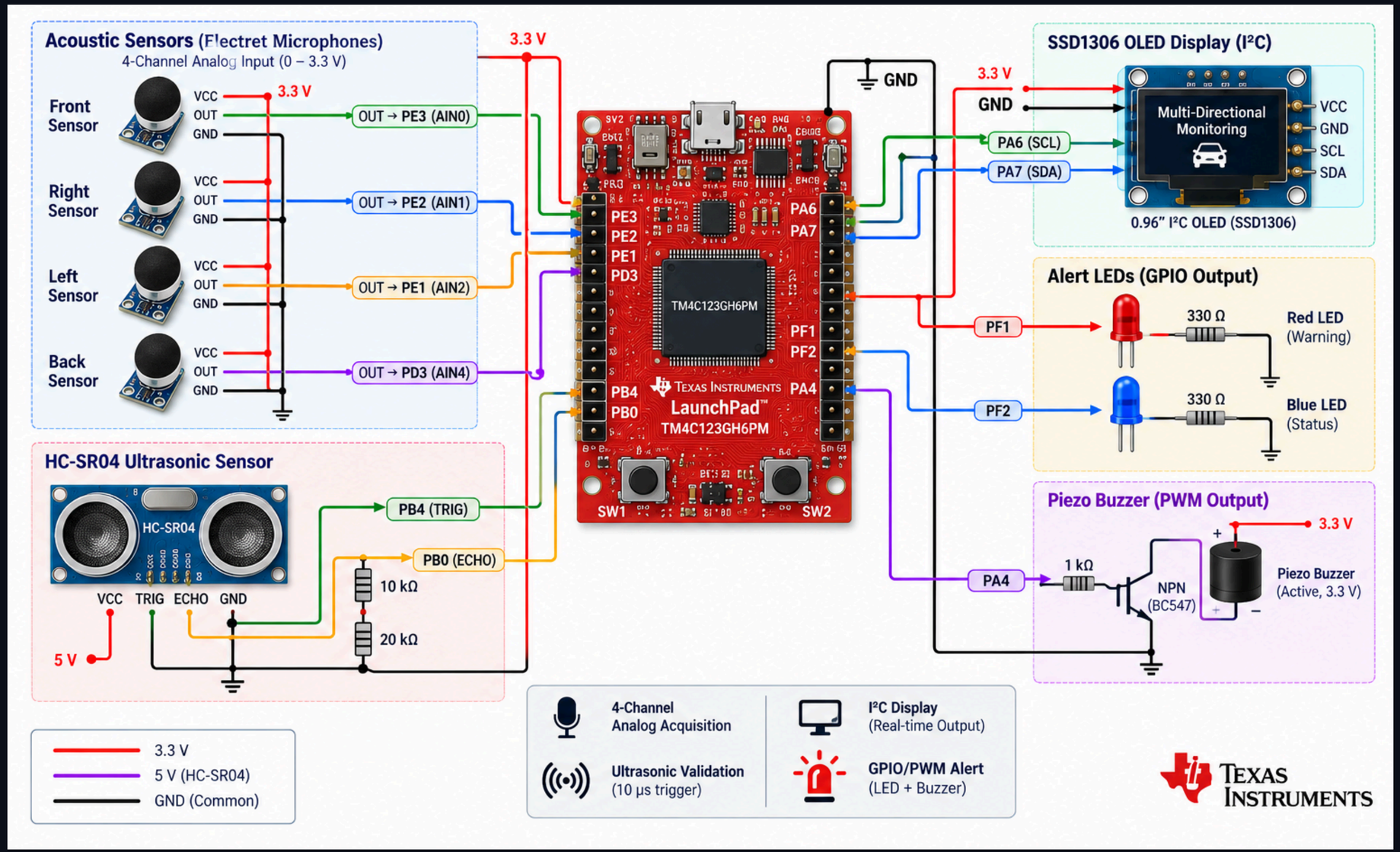
Module	TM4C123 Pin	Purpose
LED 1	PF1	Visual alert (direction)
LED 2	PF2	Visual alert (pattern)
Buzzer	PA4	Audible alert (PWM)
OLED (SCL)	PA6	I²C clock
OLED (SDA)	PA7	I²C data

WHY THESE PINS?

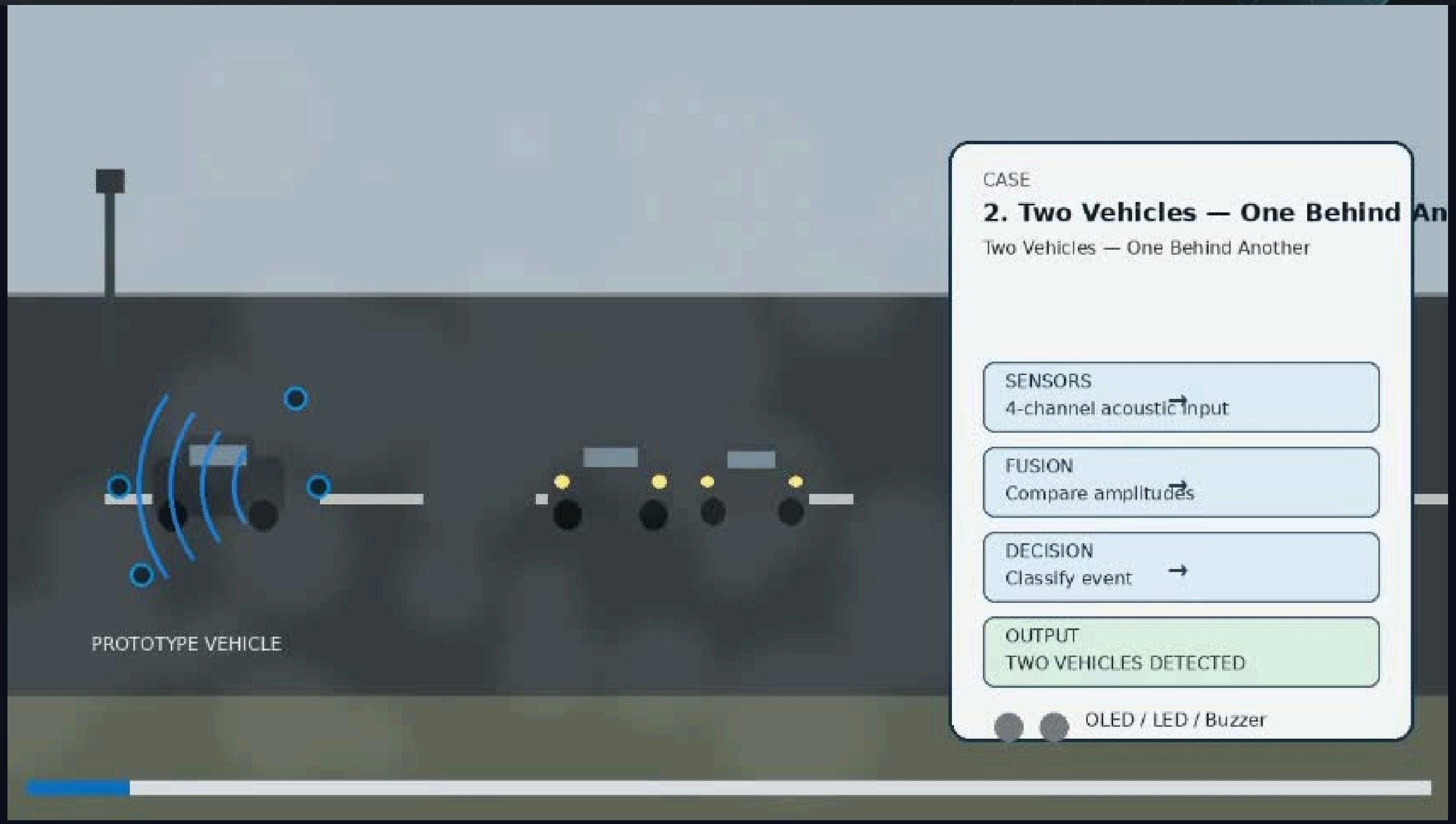
- **PE3, PE2, PE1, PD3** → ADC input channels (analog signals from 4 sensors)
- PF1, PF2 → Digital outputs for LED indication
- PA4 → PWM-capable pin for buzzer
- **PA6,** PA7 → I²C pins for OLED display (standard I²C interface)

Developed in Simulink | Validates TM4C123 implementation | Real-time direction detection | Improves road safety in low visibility

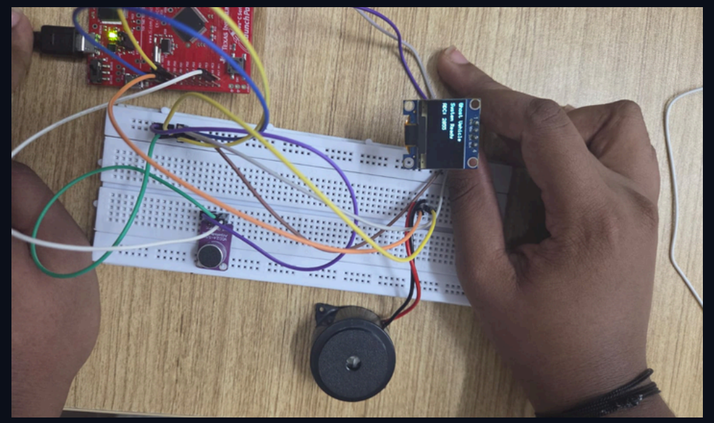
HARDWARE ARCHITECTURE & SUBSYSTEM INTERCONNECTS

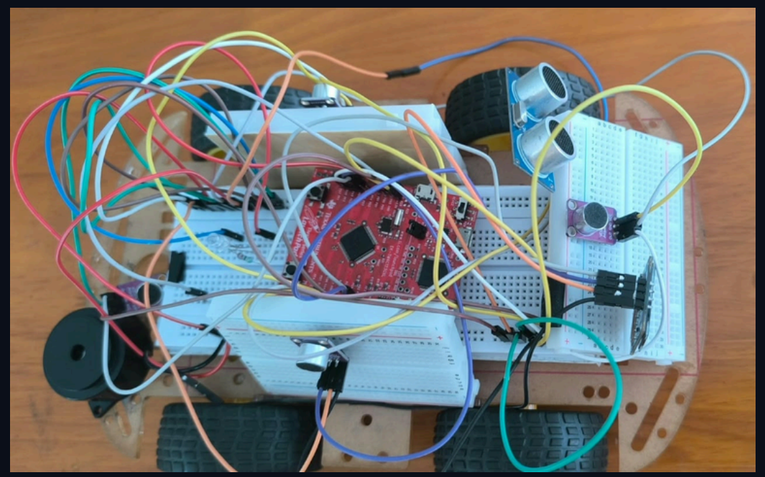


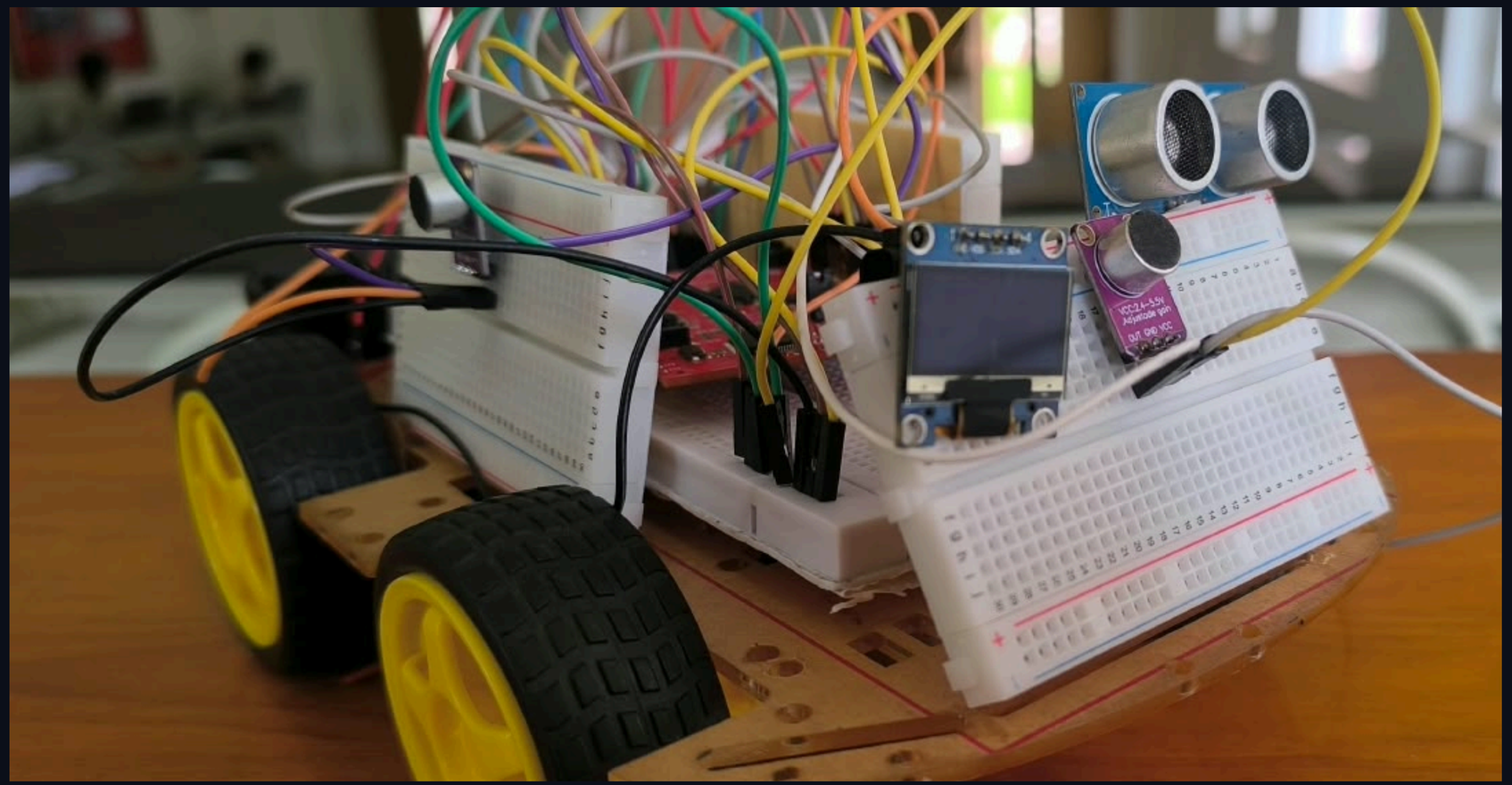
VEHICLE DETECTION & ALERT TO EACH CASE DEMO



HARDWARE DEMONSTRATION & PROTOTYPE SETUP







CONCLUSION & FUTURE WORK

CURRENT ACHIEVEMENTS
(DEMONSTRATED)

- **4 acoustic sensors**
  (front, back, left, right)
- **Real-time signal processing**
  on TM4C123
- **Detects vehicle direction**
  with clear separation
- **OLED display + LED + buzzer**
  (< 45 ms response)
- **Working prototype**
  tested in simulated fog (RC vehicle)

FUTURE WORK
(PROPOSED RESEARCD SCOPE)

- **Collect real-world data**
  (trucks, cars, two-wheelers)
- **Compare with camera & LiDAR**
  (side-by-side tests)
- **FFT / wavelet analysis**
  to filter ambient noise
- **Lightweight ML models**
  on Cortex-M4F
- **Rugged, IP67 hardware**
  for real-world roadside testing

Thank You !